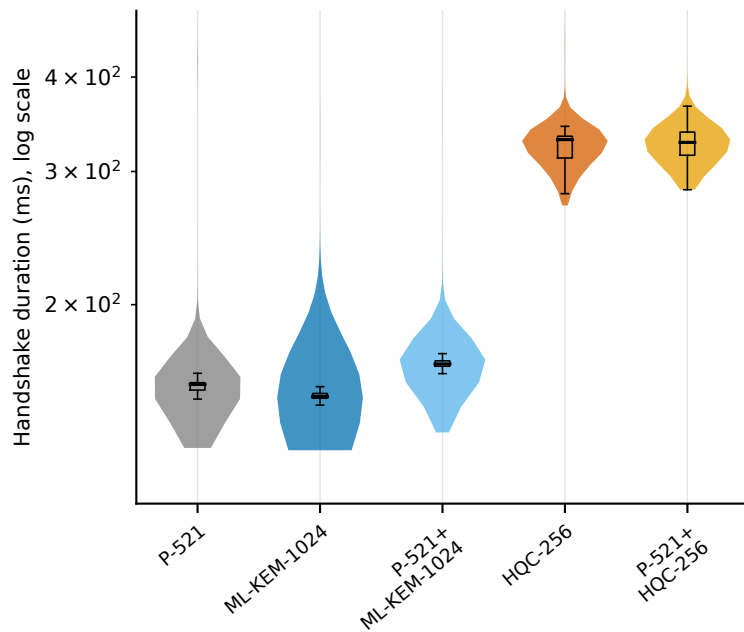
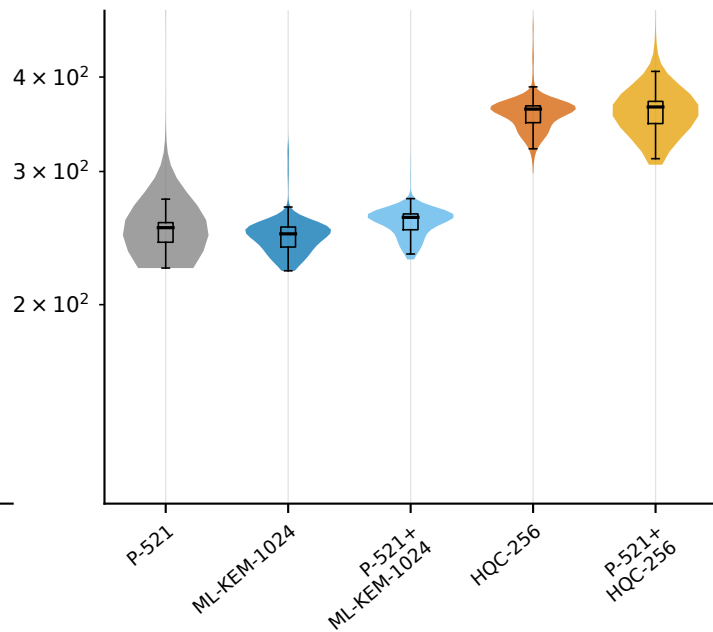


**(a)**

secp521r1 (classical)

**(b)**

mldsa87 (post-quantum)

**(c)**

Classical KEX
  ML-KEM (pure)
  ML-KEM (hybrid)
  HQC (pure)
  HQC (hybrid)

| KEM               | secp521r1<br>Mean (ms) | secp521r1<br>CV (%) | secp521r1<br>Max (ms) | mldsa87<br>Mean (ms) | mldsa87<br>CV (%) | mldsa87<br>Max (ms) |
|-------------------|------------------------|---------------------|-----------------------|----------------------|-------------------|---------------------|
| P-521             | 157.4                  | 31.8                | 1.16k                 | 257.0                | 33.9              | 1.39k               |
| ML-KEM-1024       | 159.4                  | 59.3                | 1.23k                 | 247.6                | 8.5               | 515.0               |
| P-521+ML-KEM-1024 | 169.6                  | 30.6                | 1.25k                 | 258.4                | 7.6               | 536.2               |
| HQC-256           | 327.6                  | 16.7                | 1.43k                 | 360.6                | 8.0               | 653.8               |
| P-521+HQC-256     | 329.7                  | 17.0                | 1.44k                 | 366.6                | 20.7              | 1.52k               |